

Dublin Bikes — Project Report

UCD School of Computer Science — Software Engineering Project

Kaiwen Yao • Tzuyu Chang • Ziling Huang April 2026

Authors and Contributions — each author contributed an agreed equal share of **33.33%**.

Author	%	Type of contribution
Kaiwen Yao	33.33	Code (<code>scraper</code> : standalone data collection, JCDecaux & weather ingest, Docker, Jenkins CI/CD; <code>flask-app</code> : app factory, JWT auth, station/prediction/chat APIs, DB migrations; <code>machine_learning</code> : EDA, model training, Hugging Face); Report (Process, Architecture & Design); Management (CI/CD, Docker, Scrum Master Sprint 3).
Tzuyu Chang	33.33	Code (<code>react-app</code> : architecture, routing, Tailwind, layout, maps, home, auth, profile; <code>flask-app</code> : real-time station status API); Report (Overview, Requirements, mockups, user stories); Management (User Research, Scrum Master Sprint 1 & 4).
Ziling Huang	33.33	Code (<code>flask-app</code> : weather API, journey/route planning API; <code>react-app</code> : weather component, chat page, journey planning UI); Report (Machine Learning, Testing); Management (AWS RDS, UI/UX Design, Scrum Master Sprint 2).

Screen Recording (6–8 min max). URL: *TODO* — *paste final video URL*. 3–5 min: feature walkthrough with voice, browser URL bar clearly showing the **EC2 address**. ~3 min: each member presents **1 min** on their contributions (Kaiwen → Tzuyu → Ziling).

GitHub Project. Docs/report: <https://github.com/ucdse/docs> · Backend: <https://github.com/ucdse/flask-app> · Frontend: <https://github.com/ucdse/react-app> · Scraper: <https://github.com/ucdse/scraper>

Product & Sprint Backlogs. Product backlog: *TODO* — *URL*. Sprint 1: *TODO*. Sprint 2: *TODO*. Sprint 3: *TODO*. Sprint 4: *TODO*.

Generative AI Chats (one per student, in the docs repo). KaiwenYao_Generative_AI_chats.md · TzuyuChang_Generative_AI_chats.md · ZilingHuang_Generative_AI_chats.md